AgriPower results: Farmer ownership, retail energy

Generated 28 October 2026, 17:32

Dates solved

1 January 2025, 00:00 to 31 December 2025, 23:00 (8,760 hours, taken from the hours actually solved). Year chosen: 2025.

Recommended option: best trade-off (option 4 of 10)

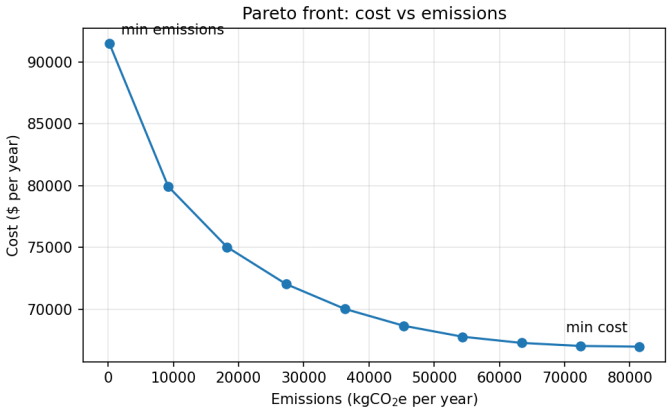
battery capacity (kWh)	840.6
battery inverter (kW)	100.0
pv capacity	383.4
generator size (kW)	0.0
cost over modelled period	72,051.1
emissions over modelled period	27,267.8
Farm bill with no assets (grid only)	101,694.6
Farm bill over period	72,051.1
Farm saving vs grid only	29,643.5
Solver status	Optimal (within the solver's gap)

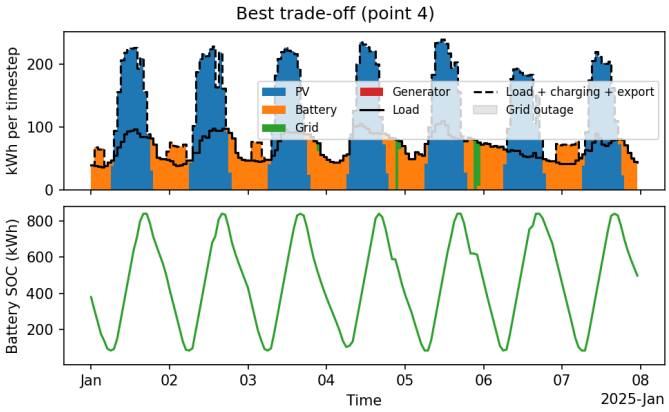
Accuracy

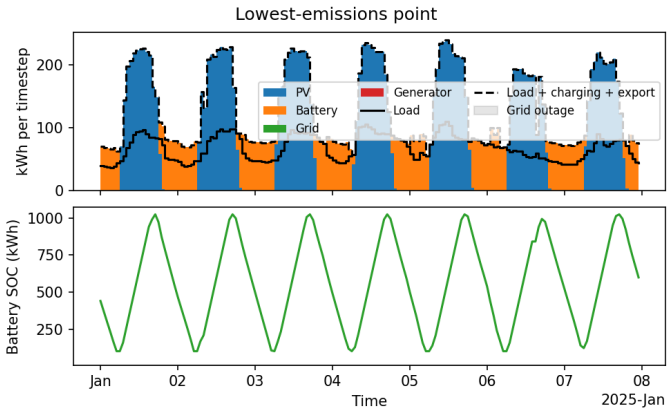
These fields are still on default values, so the optimisation result will not be fully accurate for your farm.

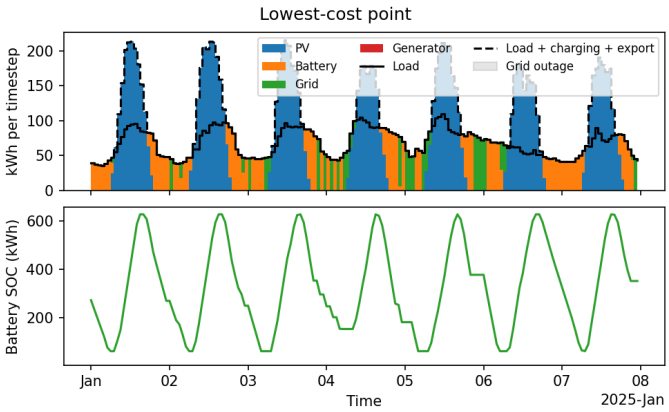
North facing roof area: 500 m2; South facing roof area: 500 m2; East facing roof area: 500 m2; West facing roof area: 500 m2; Roof tilt: Low; Farm location (latitude): -35.73; Farm location (longitude): 143.92; Grid connection (kW): 200 kW; Cost of battery ($/kWh): 232 $/kWh; Cost of solar panels ($/kWp): 1000 $/kWp; Ground array area (m2): 1000 m2; Ground array tilt (degrees): 30 degrees; Maintenance cost (% of CAPEX): 1.5%; Discount rate: 7%; Battery round trip efficiency: 95%; Battery depth of discharge: 90%; Battery lifespan (years): 15 years; Solar panel lifespan (years): 25 years.

Solver: Gurobi 13.0.3 (MIPGap 1% gap). Prices: the retail tariff in current_code/src/config.py (peak 8.789 c/kWh in 15:00 to 21:00, off-peak 6.004 c/kWh, plus its charges, exports 3 c/kWh).









Farm inputs

Field	Value	Entered or default	Group
North facing roof area	500 m2	Default	Group 1
South facing roof area	500 m2	Default	Group 1
East facing roof area	500 m2	Default	Group 1
West facing roof area	500 m2	Default	Group 1
Roof tilt	Low	Default	Group 1
Farm location (latitude)	-35.73	Default	Group 1
Farm location (longitude)	143.92	Default	Group 1
Grid connection (kW)	200 kW	Default	Group 1
Cost of battery ($/kWh)	232 $/kWh	Default	Group 1
Cost of solar panels ($/kWp)	1000 $/kWp	Default	Group 1
Ground array area (m2)	1000 m2	Default	Group 1
Ground array tilt (degrees)	30 degrees	Default	Group 1
Maintenance cost (% of CAPEX)	1.5%	Default	Group 1
Discount rate	7%	Default	Group 1
Battery round trip efficiency	95%	Default	Group 1
Battery depth of discharge	90%	Default	Group 1
Battery lifespan (years)	15 years	Default	Group 1
Solar panel lifespan (years)	25 years	Default	Group 1
Peak hours start	15:00	Default	Group 2
Peak hours end	21:00	Default	Group 2
Off-peak rate (c/kWh)	6.004 c/kWh	Default	Group 2
Peak rate (c/kWh)	8.789 c/kWh	Default	Group 2
LRET charge (c/kWh)	0	Default	Group 2
SRES charge (c/kWh)	0	Default	Group 2
VEEC charge (c/kWh)	0	Default	Group 2
Network charge (c/kWh)	0	Default	Group 2
GST	0	Default	Group 2
Other charges (c/kWh)	0	Default	Group 2
Export tariff (c/kWh)	0	Default	Group 2

Group 2 fields: Not yet used by the optimiser. A tariff scaling factor is being developed by the team.